Module 1: Introduction to cloud computing:
Private cloud, public cloud, hybrid cloud architecture.

Cloud computing is vast. It encompasses a huge range of architectural styles, classifications, and types. This complex computing network has transformed the way we work and is a crucial part of our daily lives, both at home and at work. For organizations, there are many ways to “cloud”, but let’s start with the basics of cloud computing; the internet cloud. This is generally categorized into three types:

1. Public cloud — data and other info delivered over the internet that can be shared with various people and organizations.
2. Private cloud — data and other info that is only accessible to users within your organization.
3. Hybrid cloud — a combination of the two. This environment uses public and private clouds.

Cloud computing was created because the amount of data an organization has to store and manage has grown exponentially over the past few decades. Companies were beginning to install more and more physical storage space, which became ever more expensive and cumbersome. Cloud storage removes this burden.

Your confidential data is stored in a secure, remote location. It is “the cloud” to us, but it does live in a physical location. All this means is that it is housed by a third party, not on your premises. In most cases, you don’t know where this cloud is located. You can access programs, apps, and data over the internet just as easily as if it was on your own personal computer.

The most common use cases of cloud computing include Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). In most cases, you can decide whether to set it up as a public or private option.

What Is a Public Cloud?

Anything said to live in the cloud refers to docs, apps, data, and anything else that does not reside on a physical appliance; your computer, a server, hard drive, etc.

It lives in a huge data warehouse and is accessed only over the internet. A public cloud does not mean that just anyone can log in, but it is more accessible than other types of clouds. This makes it the most popular.

A common use is wedding and vacation photos. You can upload your pictures and give an access link to all of your friends and family. Organizations of all sizes like this format because it provides:

- High scalability and elasticity

- Low-cost tier-based pricing

Public cloud services are often free or offered as a freemium or subscription-based service. The computing functionality they provide can range from basic services such as email, apps, and storage to enterprise-level OS platforms or infrastructure environments your team can use for software development and testing.

What Are the Benefits of a Public Cloud?

A “public” cloud is only accessible to people with your permission. Security is very tight. As recent history has shown, the majority of data leaks actually originate in-house. The public cloud offers:

- Strong cyber security — Attracting the most talented engineers in the world takes money. Engaging large security teams and the best security tools available is not a viable option for the average company. Cloud computing solves this problem. You benefit from having highly skilled IT professionals solely responsible for the protection of your public cloud infrastructure.
- Advanced technology creates more security innovations — More modern technology has led to advanced security services. Security innovation in the cloud is designed specifically for cloud-based solutions.
- Stringent penetration testing — Public clouds are put through stringent penetration tests and are held to stiffer standards than any on-premise solutions or private cloud options. Private clouds are regularly slack in penetration testing because it is assumed that in-house breaches are not expected.
- Controlled access — The majority of data breaches result from human error. Critics claim that keeping data in-house allows better control, but really the opposite is true. Data stored in the public cloud has fewer chances of falling into the wrong hands due to an employee’s mistake. As human control of your information decreases so does your risk.

It should be noted that cloud security is a shared responsibility. Your cloud service provider is responsible for the security of the cloud, and you are responsible for your security in-house. Customers using cloud services need to understand they play a large role in securing their data and should ensure their IT team is properly trained.

- Saves time — When you use a public cloud, you don’t have to worry about managing hosting. Your cloud solution provider is completely responsible for all of the management and maintenance of the data center. This means there is no lengthy procurement process and you don’t have to wait for operations to enable your operating system, there is no need to configure or assemble servers and you never have to worry about establishing connectivity. Additionally, the technology allows developers to employ Agile workflows, which significantly reduces lead times when it comes to testing, deploying, and releasing new products and updates.
- Saves you money — The amount of money organizations can save with the public cloud depends on the size of the operation. Large enterprises literally save

millions a year, but with improper management, cost savings may not be realized at all. Take a look at your IT environment to see how the public cloud can save you money.

- There are no capital investments — You don't need to buy expensive equipment or extra space. Public cloud subscriptions are inexpensive to set up and then you only have to pay for the resources you actually use. Your infrastructure is transformed capital expenses into more affordable operating expenses.
- Public clouds are maintained and updated by the host — You are not responsible for any additional costs. Everything is included in the cost of the subscription.
- Pay only for what you use — This eliminates paying for unused resources and you always have the flexibility to scale up or down, giving just the computing capacity you need.
- Lower energy costs — Without powering internal servers, you save money on energy costs.
- Free up IT time — IT talent can concentrate on more revenue-generating activities, instead of spending all of their time managing a data center.

Operating in the cloud is the best step forward for organizations. In addition to the most obvious benefits listed above, the cloud provides greater agility, higher efficiency, and more room to grow. When you are ahead of your competition in these areas, you can be ahead in the market, as well.

Who Are the Largest Public Cloud Providers?

The top cloud computing service providers are Amazon and Microsoft, closely followed by Google, Alibaba, and IBM. Let's take a closer look at each:

- Amazon Web Services (AWS) — AWS is an Amazon company that launched in 2002. It is currently the most popular cloud service provider in the world. It is the most comprehensive and widely adopted cloud platform that offers more than 165 full-featured services stored and provided by data centers located all over the world. Millions of customers use this service globally.
- Microsoft Azure — Microsoft Azure launched many years later than AWS and Google Cloud, but it quickly rose to the top. It is one of the fastest-growing clouds of all. Azure offers hundreds of services within various categories that include AI and Machine Learning, Compute, Analytics, Databases, DevOps, Internet of Things, and Windows Virtual Desktop.
- Google Cloud Platform (GCP) — Google's cloud is similar to AWS and Azure. It offers many of the same services in various categories, including AI and Machine Learning, computing, virtualization, storage, security, and Life Sciences. GCP services are available in 20 regions, 61 zones, and over 200 countries.
- Alibaba Cloud — Alibaba was founded in 2009. It is registered and headquartered in Singapore. The company was originally built to serve Alibaba's own e-commerce ecosystem but is now available to the public. It is the largest cloud server provider in China and offers various products and services in a wide range of categories. Alibaba is available in 19 regions and 56 zones around the world.

- International Business Machines Corporation (IBM) — IBM was founded in 1911. It is one of the oldest computer companies in the world. The cloud platform developed by IBM is built of a set of cloud computing services designed for businesses. Similar to other cloud service providers, the IBM platform includes PaaS, SaaS, and IaaS as public, private, and hybrid models.

What Is a Private Cloud?

The private cloud is a cloud solution that is dedicated to a single organization. You do not share the computing resources with anyone else. The data center resources can be located on your premises or off-site controlled by a third-party vendor. The computing resources are isolated and delivered to your organization across a secure private network that is not shared with any other customers.

The private cloud is completely customizable to meet the company's unique business and security needs. Organizations are granted greater visibility and control into the infrastructure, allowing them to operate sensitive IT workloads that meet all regulations and without compromising security or performance that could previously only be achieved with dedicated on-site data centers.

Private clouds are best suited for:

- Highly sensitive data
- Government agencies and other strictly regulated industries
- Businesses that need complete control and security over IT workloads and the underlying infrastructure
- Organizations that can afford to invest in high-performance technologies
- Large enterprises that need the power of an advanced data center able to operate efficiently and cost-effectively

What Are the Benefits of a Private Cloud?

The most common benefits of a private cloud include:

- Exclusive, dedicated environments — The private cloud is for your use only. It cannot be accessed by any other organizations.
- Somewhat scalable — The environment can be scaled as needed without tradeoffs. It is highly efficient and able to meet unpredictable demands without compromising security or performance.
- Customizable security — The private cloud complies with stringent regulations, keeping data safe and secure through protocol runs, configurations, and measures based on the company's unique workload requirements.
- Highly efficient — The performance of a private cloud is reliable and efficient.
- Flexible — The private cloud is able to transform its infrastructure according to the growing and changing needs of the organization.

Drawbacks of a Private Cloud

As effective and efficient as the private cloud may be, there are some drawbacks. These include:

- Price — The cost of a private cloud solution is quite expensive and comes with a relatively high total cost of ownership (TCO) compared to public cloud alternatives, especially in the short term.
- Not very mobile-friendly — Mobile users may have limited access to the private cloud because of the high security measures.
- Limited scalability — The infrastructure may not offer enough scalability solutions to meet all demands, especially when the cloud data center is restricted to on-site computing resources.

What Is the Difference Between a Public and Private Cloud?

A public cloud solution is all about delivering IT services directly to the client over the internet. This cloud-based service is either free, based on premiums, or by subscription according to the volume of computing resources the customer uses.

Public cloud vendors will manage, maintain, and develop the scope of computing resources shared between various customers. One main differentiating aspect of public cloud solutions is their high scalability and elasticity.

They are an affordable option with vast choices based on the requirements of the organization.

A private cloud puts the main focus on virtualization and thereby is able to separate the IT services and resources from the physical device. Applications are available virtually on the cloud, as they do not run locally on servers or end devices. It is an ideal solution for companies that deal with strict data processing and security requirements. It allocates services according to the needs of the client, making it a more flexible option.

For tighter security, a firewall is installed to protect the private cloud from any unauthorized access. Only users with security clearance are authorized to access the data on private cloud applications either by use of a secure Virtual Private Network (VPN) or over the client's intranet. Users are granted the authentication rights needed to access the services.

What Is a Hybrid Cloud?

A hybrid cloud is a computing environment that combines an on-site data center, sometimes referred to as a private cloud with the power of a public cloud. This allows the two systems to share data and applications between them as needed.

A hybrid cloud is defined as a mixed computing, storage, and services environment that is made up of a public cloud solution, private cloud services, and an on-premises infrastructure. Examples of this include Amazon Web Services (AWS) and Microsoft Azure. Using this combination gives you great flexibility and control and allows you to make the most of your infrastructure dollars.

What Are the Benefits of a Hybrid Cloud?

Although cloud services are able to save you a lot of money, their main value is in supporting an ever-changing digital business structure. Every technology management team has to focus on two main agendas: the IT side of the business and the business transformation needs. Typically, IT follows the goal of saving money. Whereas the digital business transformation side focuses on new and innovative ways of increasing revenues.

The main benefit of a hybrid cloud is its agility. To stay ahead of the game, organizations must be able to adapt and change directions as quickly as possible. This is the core principle behind every business, and a hybrid computing solution can make it work. A business might want to combine on-premises resources with private and public clouds to retain the agility needed to stay ahead in today’s world.

When computing and processing demands increase beyond what an on-premises data center can handle, businesses can tap into the cloud to instantly scale up or down to manage the changing needs. It is also a cost-effective way of getting the resources you need without spending the time or money of purchasing, installing, and maintaining new servers that you may only need occasionally.

Security Concerns of a Hybrid Solution

Hybrid cloud platforms use many of the same security measures as on-premises infrastructures, including security information and event management (SIEM). In fact, organizations that use hybrid systems find the security to be better than their on-site data center because of the automatic updates, automated data redundancy, disaster recovery, high availability, and strict cybersecurity features.

What is Multi-Cloud?

Having multiple vendors is a common practice these days. A multi-cloud architecture uses two or more cloud service providers from a number of different computing vendors. A multi-cloud environment can be several private clouds, several public clouds, or a combination of both. The main purpose of a multi-cloud environment is to reduce risks. Resources are distributed to different vendors to minimize the chance of downtime and data loss.

Multi-clouds also increase available storage and computing power.

Cloud platforms provide a variety of services and resources over the internet, allowing users to access and utilize computing power, storage, databases, networking, and other services without the need for on-premises infrastructure. There are several prominent cloud platforms, each with its unique features and strengths. As of my last knowledge update in January 2022, here are some major cloud platforms:

1. **Amazon Web Services (AWS):**
   - **Provider:** Amazon.com
   - **Key Services:** EC2 (Elastic Compute Cloud), S3 (Simple Storage Service), RDS (Relational Database Service), Lambda (Serverless Computing), and many more.
   - **Usage:** Widely adopted for various use cases, from startups to large enterprises.

2. **Microsoft Azure:**
   - **Provider:** Microsoft
   - **Key Services:** Azure Virtual Machines, Azure Blob Storage, Azure SQL Database, Azure Functions, and more.
   - **Integration:** Strong integration with Microsoft products, making it popular among enterprises using Windows-based environments.

3. **Google Cloud Platform (GCP):**
   - **Provider:** Google
   - **Key Services:** Compute Engine, Cloud Storage, Cloud SQL, Google Kubernetes Engine (GKE), and more.
   - **Strengths:** Known for its data analytics, machine learning, and container orchestration capabilities.

4. **IBM Cloud:**
   - **Provider:** IBM
   - **Key Services:** IBM Virtual Servers, IBM Cloud Object Storage, Db2 on Cloud, Watson AI services, and more.

- **Focus:** Offers a range of services, including AI and blockchain, with a focus on enterprise solutions.

5. **Alibaba Cloud:**
   - **Provider:** Alibaba Group
   - **Key Services:** Elastic Compute Service (ECS), Object Storage Service (OSS), ApsaraDB for relational and NoSQL databases, and more.
   - **Market:** Particularly dominant in the Asia-Pacific region.

6. **Oracle Cloud:**
   - **Provider:** Oracle Corporation
   - **Key Services:** Oracle Compute, Oracle Cloud Infrastructure Object Storage, Oracle Database Cloud Service, and more.
   - **Strengths:** Emphasizes on database and enterprise applications.

7. **DigitalOcean:**
   - **Provider:** DigitalOcean
   - **Key Services:** Droplets (Virtual Machines), Spaces (Object Storage), Managed Databases, and more.
   - **Usage:** Popular among developers and startups for its simplicity and developer-friendly approach.

8. **OpenStack:**
   - **Open Source:** OpenStack is an open-source cloud computing platform.
   - **Key Components:** Nova (Compute), Swift (Object Storage), Cinder (Block Storage), Keystone (Identity), Neutron (Networking), and more.
   - **Usage:** Often used for building private and hybrid clouds.

The choice of a cloud platform depends on various factors such as specific business requirements, the need for specific services, cost considerations, and integration preferences. Each platform has its strengths, and businesses may use a combination of these services based on their needs (multi-cloud strategy). Keep in mind that the cloud landscape is dynamic, and new features and services are regularly introduced by providers.